Determinants

Linear Algebra

Licenciatura em Engenharia Informática

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- A permutation of the set $\mathrm{P}_n = \{1, 2, \ldots, n\}$, $n \in \mathbb{N}$, is a bijection from P_n to P_n.
- The set of all bijections, σ_i, P_n $i = 1, 2, \ldots, n!$, is denoted by Σ_n.
- The inversion number, $\mathrm{N}(\sigma)$, of a permutation $\sigma \in \Sigma_n$, is the number of pairs $(x, y) \in \mathrm{P}_n^2$ that satisfy the condition

$$x < y \quad \text{and} \quad \sigma(x) > \sigma(y).$$

Definição 1 (Determinant of order n)

Let n be a positive integer. The determinant of order n is the function

$$\det : \mathscr{M}_n \longrightarrow \mathbb{R}$$

such that, given a matrix $\mathrm{A} = [a_{i,j}] \in \mathscr{M}_n$ we have

$$\det(\mathrm{A}) = \sum_{i=1}^{n!} (-1)^{N(\sigma_i)} a_{1,\sigma_i(1)} a_{2,\sigma_i(2)} \cdots a_{n,\sigma_i(n)} \tag{1}$$



- ***Usually we do not compute determinants using the definition formula 1.***
- *The calculation of determinants of high order matrices, using the equality (1), implies a huge calculation effort. For example to compute the determinant of a matrix of order* 15 *we need to perform* $15! = 1307674368000$ *sums and each term has* 15 *factors...*
- *To overcome this problem we will begin to find how to compute lower order determinants (determinants of* 1*st,* 2 *th e* 3 *th order) and find algorithms that calculate higher-order determinants with reduced effort.*

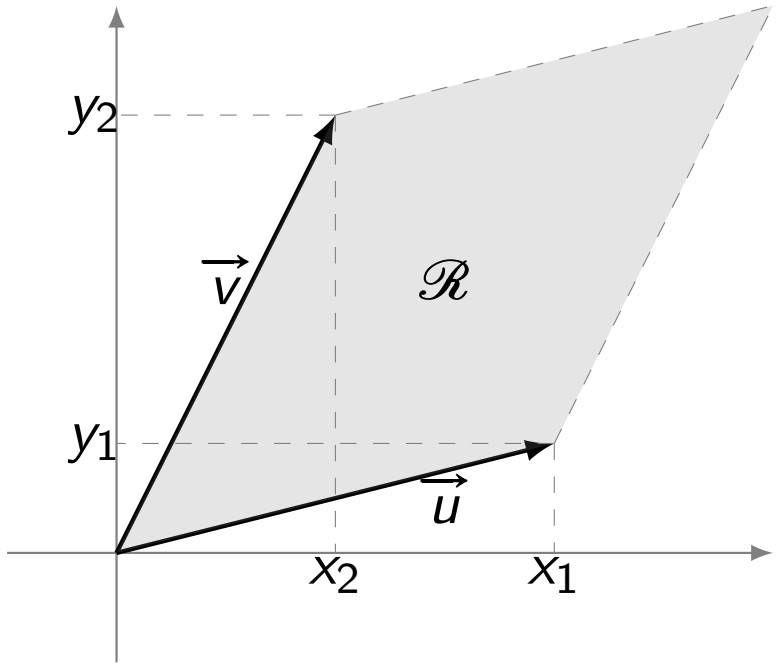
Property 1 (determinants of order 1 and 2)

1. The determinant of a matrix $A = [a_{1,1}] \in \mathscr{M}_1$ is equal to it entry. I. e. $\det(A) = a_{1,1}$.
2. The determinant of a matrix $A = [a_{i,j}] \in \mathscr{M}_2$ is computed by the following relation

$$\det(A) = \begin{vmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{vmatrix} = a_{1,1}a_{2,2} - a_{1,2}a_{2,1}.$$

☞ *O determinant of matrix* A *is the product of the main diagonal minus the product of the elements of the secondary diagonal.*

Geometric interpretation: We can interpret geometrically the absolute value of a determinant of order 2 as the area of a parallelogram defined by two vectors, $\vec{u} = (x_1, y_1)$ and $\vec{v} = (x_2, y_2)$ in $\mathbb{R}^2$.



$$\text{Area}(\mathscr{R}) = \left|\det\left(\begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix}\right)\right|$$

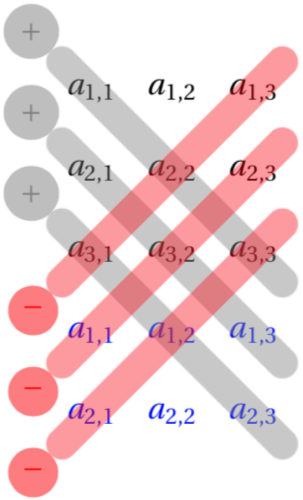
Property 2 (Determinant of order 3)

Consider the matrix $A = [a_{i,j}] \in \mathscr{M}_3$. Then

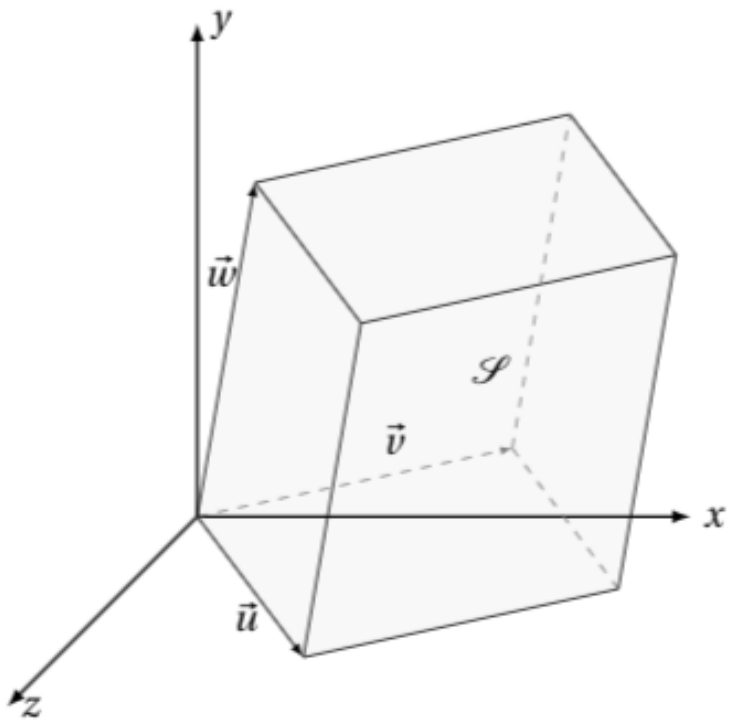
$$\det(A) = \begin{vmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{vmatrix} = \left(a_{1,1}a_{2,2}a_{3,3} + a_{2,1}a_{3,2}a_{1,3} + a_{3,1}a_{1,2}a_{2,3}\right) -$$

$$\left(a_{1,3}a_{2,2}a_{3,1} + a_{2,3}a_{3,2}a_{1,1} + a_{3,3}a_{1,2}a_{2,1}\right)$$

☞ *Mnemonic, known as* ***Sarrus' Rule****,to memorize the formula for calculating a determinant of order* 3.



Geometric interpretation: We can geometrically interpret the absolute value of the determinant of order 3, as the volume of a parallelepiped defined by three vectors $\vec{u} = (x_1, y_1, z_1)$, $\vec{v} = (x_2, y_2, z_2)$ and $\vec{w} = (x_3, y_3, z_3)$ in $\mathbb{R}^3$.



$$\text{Volume}(\mathscr{S}) = \left| \det\left(\begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix} \right) \right|$$

Properties 1

Consider matrix $A = [a_{i,j}] \in \mathscr{M}_n$. Then we have:

1. $\det(A) = \det(A^T)$.
2. If A is a lower triangular or upper triangular matrix then,

$$\det(A) = a_{1,1}a_{2,2}\ldots a_{n,n}.$$



- *Diagonal matrices are special cases of triangular matrices, and their determinant is the product of the entries on the main diagonal.*
- $det(I_n) = 1$.

Properties 2 (alternating multilinear function)

Seja $A = [L_1, L_2, \ldots, L_n] \in \mathscr{M}_n$. A determinant function n,

$$\det : \mathscr{M}_n \longrightarrow \mathbb{R},$$

is a row n-**linear** because, for all row i we have

$$\det([L_1, \ldots, L_{i-1}, \underbrace{\alpha L_i' + \beta L_i''}_{L_i}, L_{i+1}, \ldots, L_n]) =$$

$$\alpha \det([L_1, \ldots, L_{i-1}, L_i', L_{i+1}, \ldots, L_n]) + \beta \det([L_1, \ldots, L_{i-1}, L_i'', L_{i+1}, \ldots, L_n])$$

for all i and scalars $\alpha, \beta \in \mathbb{R}$.
On the other hand, it is said that the determinant function is **row alternating multilinear function** because it satisfies the equality

$$\det([L_1, \ldots, L_i, \ldots, L_j, \ldots, L_n]) = -\det([L_1, \ldots, L_j, \ldots, L_i, \ldots, L_n])$$

In other words, if we swap two rows i and j, with $i \neq j$, then the resulting matrix is the negative of the determinant of the original matrix.

As particular cases of n-linearity, we have that:

$$\det([L_1,\ldots,L_{i-1},\alpha L_i,L_{i+1},\ldots,L_n]) = \alpha\det([L_1\ldots,L_{i-1},L_i,L_{i+1},\ldots,L_n])$$

$$\det([L_1,\ldots,L_{i-1},\underbrace{L_i'+L_i''}_{L_i},L_{i+1},\ldots,L_n]) =$$

$$\det([L_1,\ldots,L_{i-1},L_i',L_{i+1},\ldots,L_n]) + \det([L_1,\ldots,L_{i-1},L_i'',L_{i+1},\ldots,L_n])$$



As, for all matrices $A \in \mathscr{M}_n$, $\det(A) = \det(A^T)$ then we have that the determinant function is also a n-linear and alternating function with respect to the columns..

Examples:

a)

$$\begin{vmatrix} a & b & c & d+2e \\ f & g & h & i+2j \\ l & m & n & o+2p \\ q & r & s & t+2e \end{vmatrix} = \begin{vmatrix} a & b & c & d \\ f & g & h & i \\ l & m & n & o \\ q & r & s & t \end{vmatrix} + 2 \begin{vmatrix} a & b & c & e \\ f & g & h & j \\ l & m & n & p \\ q & r & s & e \end{vmatrix}$$

b)

$$\begin{vmatrix} a & b & 0\times c & d \\ e & f & 0\times g & h \\ i & j & 0\times l & m \\ n & o & 0\times p & q \end{vmatrix} = 0 \times \begin{vmatrix} a & b & c & d \\ e & f & g & h \\ i & j & l & m \\ n & o & p & q \end{vmatrix} = 0$$

c)

$$\begin{vmatrix} a & b & c & d \\ e & f & g & h \\ \lambda a & \lambda b & \lambda c & \lambda d \\ i & j & k & l \end{vmatrix} = \lambda \begin{vmatrix} a & b & c & d \\ e & f & g & h \\ a & b & c & d \\ i & j & k & l \end{vmatrix} = -\lambda \begin{vmatrix} a & b & c & d \\ e & f & g & h \\ a & b & c & d \\ i & j & k & l \end{vmatrix} = 0$$

Properties 3

Let $A \in \mathscr{M}_n$ be a matrix. Then we have

1. If A has a zero row or column, then its determinant is zero.
2. If A has two identical rows or two identical columns, then its determinant is zero.
3. If A has a row that is a multiple of another row or a column that is a multiple of another column, then its determinant is zero.

The key idea of using elementary operations in the calculation of determinants, of any order, is to transform a matrix into a triangular matrix.

Property 3 (As determinant is an alternating function we ha

$$A \xrightarrow{L_i \leftrightarrow L_j} A', \quad i \neq j, \quad \Longrightarrow \quad \det(A') = -\det(A)$$

and,

$$A \xrightarrow{C_i \leftrightarrow C_j} A', \quad i \neq j, \quad \Longrightarrow \quad \det(A') = -\det(A)$$

That is, whenever we perform a row or column swap, we must change the sign of the determinant..

Property 4 (As determinant is a n-linear function we have)

$$A \xrightarrow{L_i \to \alpha L_i} A' \quad \Longrightarrow \quad \det(A') = \alpha \det(A), \quad \forall \alpha \in \mathbb{R}$$

and,

$$A \xrightarrow{C_i \to \alpha C_i} A' \quad \Longrightarrow \quad \det(A') = \alpha \det(A), \quad \forall \alpha \in \mathbb{R}$$

That is, if we multiply a row or a column of a matrix by a scalar α, the determinant of the resulting matrix is the determinant of the original matrix multiplied by α.

Property 5

$$A \xrightarrow{L_i \leftarrow L_i + \alpha L_j} A' \implies \det(A') = \det(A), \quad \forall \alpha \in \mathbb{R}$$

e,

$$A \xrightarrow{C_i \leftarrow C_i + \alpha C_j} A' \implies \det(A') = \det(A), \quad \forall \alpha \in \mathbb{R}$$

That is, if we add a multiple of one row (or column) to another row (or column), the determinant does not change.

Example 1:

$$\begin{vmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & -4 & 0 & 0 \\ -5 & -4 & 0 & 0 \end{vmatrix} \overset{L_1 \leftrightarrow L_4}{=\!=} - \begin{vmatrix} -5 & -4 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{vmatrix} \overset{L_2 \leftrightarrow L_3}{=\!=} \begin{vmatrix} -5 & -4 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{vmatrix} = 120$$

Example 2:

$$\begin{vmatrix} 2 & -2 & 4 & 6 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \\ 5 & -4 & 10 & 14 \end{vmatrix} \overset{L_1 \leftarrow \frac{1}{2} L_1}{=\!=} 2 \begin{vmatrix} 1 & -1 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \\ 5 & -4 & 10 & 14 \end{vmatrix} \overset{L_4 \leftarrow L_4 - 5L_1}{=\!=} 2 \begin{vmatrix} 1 & -1 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \end{vmatrix} =$$

$$\overset{L_2 \leftrightarrow L_4}{=\!=} -2 \begin{vmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{vmatrix} \overset{L_3 \leftrightarrow L_4}{=\!=} 2 \begin{vmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \end{vmatrix} = 2 \times 3 = 6$$

Example 3:

$$\begin{vmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 1 \end{vmatrix} \overset{C_1 \leftarrow C_1 - 2C_5}{=\!=} \begin{vmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{vmatrix} \overset{C_2 \leftrightarrow C_4}{=\!=} - \begin{vmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{vmatrix} = -1$$

Properties 4

Consider the matrices A and $B \in \mathscr{M}_n$. Then

1. A is invertible if and only if $\det(A) \neq 0$.
2. $\det(AB) = \det(A)\det(B)$
3. $\det(\alpha A) = \alpha^n \det(A)$, $\forall \alpha \in \mathbb{R}$
4. If A is invertible then,

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$



- *Another way to calculate the determinant is through* ***teorema de Laplace****.*
- *Laplace's theorem reduces the calculation of an n-order determinant to the calculation of n determinants of* $(n-1)$*-order.*

Definição 2 (Cofactor (of $a_{k,\ell}$))

Let $A=[a_{i,j}] \in \mathscr{M}_n$ be a matrix. The *cofactor* of an entry $a_{k,\ell}$ of A is the determinant of the matrix of order $n-1$ obtained from A by deletion of the row k and column ℓ . The cofactor $a_{k,\ell}$ é representado por $m_{k,\ell}$.

Definição 3 (Algebraic cofactor (of $a_{i,j}$))

Let $A=[a_{i,j}] \in \mathscr{M}_n$ be a matrix. The algebraic cofactor, $A_{k,\ell}$ of entry $a_{k,\ell}$ of A is defined by

$$A_{k,\ell} = (-1)^{k+\ell} m_{k,\ell}.$$

Nota: We have $|A_{k,\ell}| = |m_{k,\ell}|$. If $k+\ell$ is a even number we have $A_{k,\ell} = m_{k,\ell}$ and $A_{k,\ell} = -m_{k,\ell}$ if $k+\ell$ is odd.

Example:
Given the matrix

$$A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -1 & -3 \\ 3 & 1 & 0 & -3 \end{bmatrix}.$$

We have,

$$m_{1,1} = \begin{vmatrix} 1 & 1 & -1 \\ 0 & -1 & -3 \\ 1 & 0 & -3 \end{vmatrix} = -1; \quad m_{2,3} = \begin{vmatrix} 1 & 0 & 2 \\ 0 & 0 & -3 \\ 3 & 1 & -3 \end{vmatrix} = 3; \quad m_{3,4} = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 3 & 1 & 0 \end{vmatrix} = -4;$$

$$m_{2,2} = \begin{vmatrix} 1 & 1 & 2 \\ 0 & -1 & -3 \\ 3 & 0 & -3 \end{vmatrix} = 0; \quad m_{3,1} = \begin{vmatrix} 0 & 1 & 2 \\ 1 & 1 & -1 \\ 1 & 0 & -3 \end{vmatrix} = 0; \quad m_{4,4} = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{vmatrix} = -1.$$

and

$$A_{1,1} = (-1)^{1+1} m_{1,1} = -1; \quad A_{2,3} = (-1)^{2+3} m_{2,3} = -3; \quad A_{3,4} = (-1)^{3+4} m_{3,4} = 4;$$

$$A_{2,2} = (-1)^{2+2} m_{2,2} = 0; \quad A_{3,1} = (-1)^{3+1} m_{3,1} = 0; \quad A_{4,4} = (-1)^{4+4} m_{3,4} = -1.$$

Teorema 1 (Laplace)

Consider the matrix $A = [a_{i,j}] \in \mathscr{M}_n$ and the positive integers k, ℓ tais que $1 \leq k, \ell \leq n$. Then we have

$$\begin{aligned} \det(A) &= \sum_{i=1}^{n} a_{k,i} A_{k,i} \quad \text{Laplace expansion by row } k \\ &= \sum_{i=1}^{n} a_{i,\ell} A_{i,\ell} \quad \text{Laplace expansion by column } \ell \end{aligned}$$



- *We can calculate the Laplace expansion using any row or any column.*
- *In practice, it is convenient to choose, if possible, a row or column that has a large number of zero entries..*
- *One can apply elementary operations to the matrix to zero out entries in a particular row (or column) and then expand along that row (or column).*

Example: Computing the determinant of

$$A = \begin{bmatrix} 1 & -1 & 3 & 0 \\ -4 & 0 & 0 & 1 \\ -2 & 1 & 1 & -1 \\ 0 & -3 & 2 & 1 \end{bmatrix}$$

using Laplace's theorem.
We can choose any row for the Laplace expansion. However, for calculation purposes, it is convenient to choose develop the 2nd row. Then we have,

$$\det(A) = a_{2,1}\,A_{2,1} + \cancel{a_{2,2}\,A_{2,2}} + \cancel{a_{2,3}\,A_{2,3}} + a_{2,4}\,A_{2,4}$$

As $a_{2,2} = a_{2,3} = 0$ we save the calculation of two algebraic cofactors, $A_{2,2}$ e $A_{2,3}$. and the Laplace expansion along the 2nd row

$$\det(A) = (-4) \times (-1)^{2+1} \begin{vmatrix} -1 & 3 & 0 \\ 1 & 1 & -1 \\ -3 & 2 & 1 \end{vmatrix} + 1 \times (-1)^{2+4} \times \begin{vmatrix} 1 & -1 & 3 \\ -2 & 1 & 1 \\ 0 & -3 & 2 \end{vmatrix}.$$

Using, for example, Sarrus' rule we have

$$\begin{vmatrix} -1 & 3 & 0 \\ 1 & 1 & -1 \\ -3 & 2 & 1 \end{vmatrix} = 3; \qquad \begin{vmatrix} 1 & -1 & 3 \\ -2 & 1 & 1 \\ 0 & -3 & 2 \end{vmatrix} = 19$$

Then, we have $\det(A) = 4 \times 3 + 1 \times 19 = 31$

Example (cont.) On the other hand, one can combine the properties of determinants with the Laplace expansion to calculate the determinant of a matrix. We can set one of the entries in the 2nd row of the matrix A to zero. In fact, we have:

$$\det(A) = \begin{vmatrix} 1 & -1 & 3 & 0 \\ -4 & 0 & 0 & 1 \\ -2 & 1 & 1 & -1 \\ 0 & -3 & 2 & 1 \end{vmatrix} \overset{C_1 \leftarrow C_1 + 4C_4}{=\!=\!=} \begin{vmatrix} 1 & -1 & 3 & 0 \\ 0 & 0 & 0 & 1 \\ -6 & 1 & 1 & -1 \\ 4 & -3 & 2 & 1 \end{vmatrix}$$

Performing the Laplace expansion along the second row, we obtain:

$$\det(A) = \cancel{a'_{2,1} A'_{2,1}} + \cancel{a'_{2,2} A'_{2,2}} + \cancel{a'_{2,3} A'_{2,3}} + a'_{2,4} A'_{2,4}$$

then

$$\det(A) = 1 \times (-1)^{2+4} \begin{vmatrix} 1 & -1 & 3 \\ -6 & 1 & 1 \\ 4 & -3 & 2 \end{vmatrix} = 31.$$



Calculating the inverse of a matrix using the adjoint matrix

Definição 4 (Adjoint matrix)

Let $A \in \mathscr{M}_n$ be a matrix. The adjoint matrix of A, adj(A), is the the transpose of the matrix obtained from A by replacing its elements with theis respective algebraic cofactors.

Example: Given the matrix

$$A = \begin{bmatrix} 1 & -1 \\ -3 & 2 \end{bmatrix}$$

we have

$$\text{Adj}(A) = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{1,2} \end{bmatrix}^T = \begin{bmatrix} (-1)^2 \times 2 & (-1)^3 \times (-3) \\ (-1)^3 \times (-1) & (-1)^4 \times 1 \end{bmatrix}^T = \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix}.$$

Teorema 2

Let $A \in \mathscr{M}_n$. then:

1. Tem-se
$$A \cdot \text{Adj}(A) = \text{Adj}(A) \cdot A = \det(A) \cdot I_n$$
2. If A is invertible then,
$$A^{-1} = \frac{1}{\det(A)} \cdot \text{Adj}(A)$$

In the previous example, we have $\det(A) = -1$, then

$$\begin{bmatrix} 1 & -1 \\ -3 & 2 \end{bmatrix}^{-1} = \frac{1}{-1} \cdot \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ -3 & -1 \end{bmatrix}$$